

Port Scan Activity [Let's Defend – Investigate]

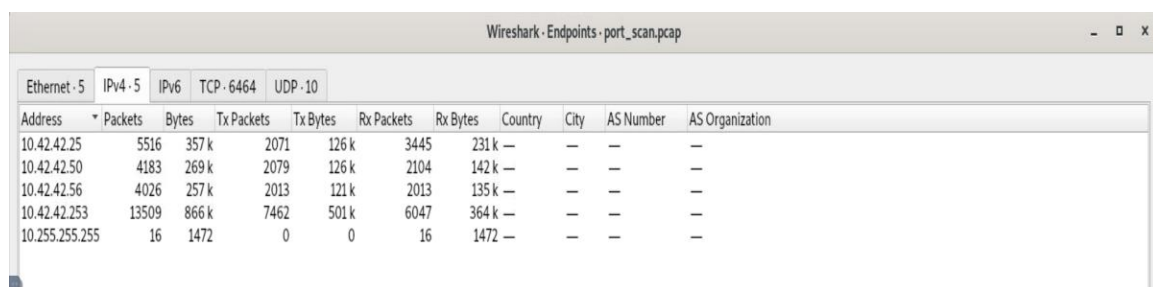
Can you determine evidences of port scan activity?

Log file: /root/Desktop/ChallengeFile/port_scan.pcap

Note: pcap file found public resources.

| Start Investigation

Q1: What is the IP address scanning the environment?

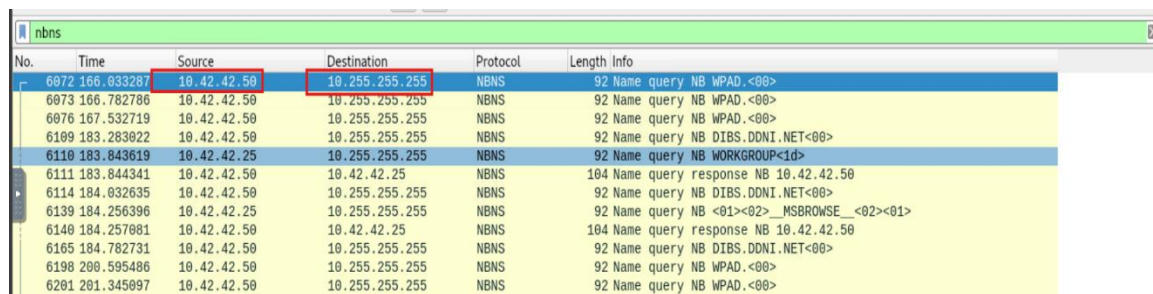


Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	AS Number	AS Organization
10.42.42.25	5516	357 k	2071	126 k	3445	231 k	—	—	—	—
10.42.42.50	4183	269 k	2079	126 k	2104	142 k	—	—	—	—
10.42.42.56	4026	257 k	2013	121 k	2013	135 k	—	—	—	—
10.42.42.253	13509	866 k	7462	501 k	6047	364 k	—	—	—	—
10.255.255.255	16	1472	0	0	16	1472	—	—	—	—

The scanning IP address is 10.42.42.253. I identified it because it has the highest number of transmitted packets (13,509 packets) and the largest amount of data traffic compared to other hosts in the network.

Answer: 10.42.42.253

Q2: What is the IP address found as a result of the scan?



No.	Time	Source	Destination	Protocol	Length	Info
6072	166.033287	10.42.42.50	10.255.255.255	NBNS	92	Name query NB WPAD.<00>
6073	166.782786	10.42.42.50	10.255.255.255	NBNS	92	Name query NB WPAD.<00>
6076	167.532719	10.42.42.50	10.255.255.255	NBNS	92	Name query NB WPAD.<00>
6109	183.283022	10.42.42.50	10.255.255.255	NBNS	92	Name query NB DIBS.DDNI.NET<00>
6110	183.843619	10.42.42.25	10.255.255.255	NBNS	92	Name query NB WORKGROUP<1d>
6111	183.844341	10.42.42.50	10.42.42.25	NBNS	104	Name query response NB 10.42.42.50
6114	184.032635	10.42.42.50	10.255.255.255	NBNS	92	Name query NB DIBS.DDNI.NET<00>
6139	184.256396	10.42.42.25	10.255.255.255	NBNS	92	Name query NB <01><02>_MSBROWSE_<02><01>
6140	184.257081	10.42.42.50	10.42.42.25	NBNS	104	Name query response NB 10.42.42.50
6165	184.782731	10.42.42.50	10.255.255.255	NBNS	92	Name query NB DIBS.DDNI.NET<00>
6198	200.595486	10.42.42.50	10.255.255.255	NBNS	92	Name query NB WPAD.<00>
6201	201.345097	10.42.42.50	10.255.255.255	NBNS	92	Name query NB WPAD.<00>

The scanning process revealed the IP address 10.42.42.50. This was confirmed by observing multiple NBNS name queries sent from this host to the broadcast address (10.255.255.255) to discover other devices on the network.

Answer: 10.42.42.50

Q3: What is the MAC address of the Apple system it finds?

Head to **Statistics** → **Conversations** → **Ethernet** tab and toggle the **Name Resolution** option at the bottom. We are able to find a Apple system.

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
Apple_92:6e:dc	QuantaCo_82:1f:4a	5412	348 k	2007	120 k	3405	228 k	0.607596	603.2091	1600	3024
Apple_92:6e:dc	Broadcast	4	368	4	368	0	0	183.843619	360.4177	8	0
Apple_92:6e:dc	Compalln_51:d7:b2	100	7964	60	4968	40	2996	183.844341	360.7820	110	66
QuantaCo_82:1f:4a	Compalln_51:d7:b2	4071	259 k	2044	137 k	2027	122 k	0.000000	603.9423	1823	1619
QuantaCo_82:1f:4a	Compalln_cb:1e:79	4026	257 k	2013	135 k	2013	121 k	0.607594	605.4745	1795	1610
Compalln_51:d7:b2	Broadcast	12	1104	12	1104	0	0	166.033287	53.3117	165	0

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
00:16:cb:92:6e:dc	00:23:8b:82:1f:4a	5412	348 k	2007	120 k	3405	228 k	0.607596	603.2091	1600	3024
00:16:cb:92:6e:dc	ff:ff:ff:ff:ff:ff	4	368	4	368	0	0	183.843619	360.4177	8	0
00:16:cb:92:6e:dc	70:5a:b6:51:d7:b2	100	7964	60	4968	40	2996	183.844341	360.7820	110	66
00:23:8b:82:1f:4a	70:5a:b6:51:d7:b2	4071	259 k	2044	137 k	2027	122 k	0.000000	603.9423	1823	1619
00:23:8b:82:1f:4a	00:26:22:cb:1e:79	4026	257 k	2013	135 k	2013	121 k	0.607594	605.4745	1795	1610
70:5a:b6:51:d7:b2	ff:ff:ff:ff:ff:ff	12	1104	12	1104	0	0	166.033287	53.3117	165	0

Answer: 00:16:cb:92:6e:dc

Q4: What is the IP address of the detected Windows system?

Head to *Statistics* → *Endpoint* → *IPv4*

From what we gather so far:

- 10.42.42.25 – Apple System
- 10.42.42.50 – Windows System
- 10.42.42.56 – Unknown System
- 10.42.42.253 – Attacking System
- 10.255.255.255 – Network Broadcast Address

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.42.42.25	10.42.42.253	5412	348 k	2007	120 k	3405	228 k	0.607596	603.2091	1600	3024
10.42.42.25	10.255.255.255	4	368	4	368	0	0	183.843619	360.4177	8	0
10.42.42.25	10.42.42.50	100	7964	60	4968	40	2996	183.844341	360.7820	110	66
10.42.42.25	10.42.42.253	4071	259 k	2027	122 k	2044	137 k	0.000000	603.9423	1619	1823
10.42.42.50	10.255.255.255	12	1104	12	1104	0	0	166.033287	53.3117	165	0
10.42.42.56	10.42.42.253	4026	257 k	2013	121 k	2013	135 k	0.607594	605.4745	1795	1610

Answer: 10.42.42.50